

North Sky As-Built Tool

Complete Technical Specification

North Sky Communications | Lumen / CenturyLink Contract Work

Developer: William Keesee | Version: Production — May 2026

1. Application Overview

The North Sky As-Built Tool is a browser-based field documentation platform built for telecom construction foremen and supervisors working under the Lumen/CenturyLink contract. Its core purpose is to let crew leads draw as-built markups of completed work on a digital canvas — and simultaneously auto-generate the exact Lumen billing unit codes, quantities, and units of measure required for invoice submission.

The app eliminates manual billing lookups. Every tool placed on the canvas is backed by a billing engine (units.js) that resolves the correct Lumen unit codes based on the attributes the foreman enters in each tool's modal popup.

Core Workflow

- Foreman enters a Job Number → app looks up the work order in Smartsheet → auto-populates address, supervisor, foreman, notes
- Google Map of the job site is captured as a satellite background image on the canvas
- Foreman selects tools from the sidebar (Pole, HH, Fiber Cable, Trench, etc.) and places them on the canvas
- Each tool triggers a premium dark-glass modal with smart attribute fields
- On confirm → object renders on canvas with correct color/label + billing units auto-populate in the floating Billable Units panel
- Canvas exports to PDF for submission

2. File Structure

File	Purpose
index.html	Main shell — header, tab nav, modals, canvas, scripts
style.css	All styles (~2,188 lines)
js/app.js	Fabric.js canvas engine + all interactive logic (~4,224 lines)
js/units.js	Smart Unit Dictionary + billing resolver (~1,574 lines)
js/contract_units.json	746 raw Lumen unit codes (reference lookup)
assets/logo.jpg	North Sky Communications logo
ss-worker.js	Cloudflare Worker — Smartsheet API proxy
wrangler.toml	Cloudflare deployment configuration

3. Technology Stack

Layer	Technology
Canvas Engine	Fabric.js 5.3.0 (CDN)
PDF Export	jsPDF 2.5.1 + html2canvas 1.4.1
Map	Google Maps JS API + Places Library

Layer	Technology
Fonts	Rajdhani (modal titles) + Inter (UI) via Google Fonts CDN
Frontend	Vanilla JS ES Modules — no framework
Hosting	Vercel (static file hosting)
Backend Proxy	Cloudflare Worker: nsc-smartapp.williamkeesee06.workers.dev
Job Data	Smartsheet API (Sheet ID: 1833739362822020)
Maps API Key	AlzaSyBRrs_7R-edB7ffraBtkSAWmdSa3OT0XuA

4. Color Convention (Never Change)

These colors are hardcoded into every canvas object based on the work status:

Color	Hex	Usage
NEW	#FF0000	Red — new installations
EXISTING	#000000	Black — existing infrastructure (label only, no billing)
REMOVE	#00AA00	Green — removals
XFER	#0000FF	Blue — transfers / rearrangements
DE_RELASH	#A020F0	Purple — de-lash / re-lash
TRENCH / BORE / POTHOLE	#8B4513	Brown — all underground civil work (hardcoded)

5. Layout Architecture

Zone	Description
App Header (fixed)	Logo, Job# + Lookup, Address/City, Zoom buttons (- + <>>), Save, Login
Notes Bar (fixed)	Supervisor, Foreman, Date, Job Status, TC Required
Style Strip (fixed)	COLOR ▪ STROKE ▪ STYLE OBJ COLOR / STROKE / FONT SIZE GROUP / UNGROUP
Tab Strip (48px sidebar)	Aerial, UG, Cable, Splicing, XFers, Misc, Tools — vertical icon tabs
Tool Panel (220px)	Expandable sidebar with tool buttons per tab. Auto-opens to Tools on load.
Canvas (Fabric.js)	Full remaining width/height. Dot-grid background. Scroll=pan, Ctrl+scroll=zoom.
Billable Units Panel	Floating, draggable, minimizable panel — bottom-right default position

CSS Variables (auto-calculated by `sizeCanvas()`)

- `--header-h` — actual rendered header height
- `--notes-h` — notes bar height
- `--strip-h` — style strip height

- `canvas-area top = header-h + notes-h + strip-h`

6. Billing Engine (units.js)

The core of the app. Every tool's `mapping(input)` function takes the modal's collected values and returns a raw result with canvas styling + billing unit array.

Raw Result Shape (returned by mapping function)

Field	Type	Description
<code>color</code>	string	Hex color for the canvas object
<code>label</code>	string?	Optional text label rendered on canvas
<code>units[]</code>	array	Array of { <code>code</code> , <code>type</code> , <code>qty</code> , <code>unit</code> } billing line items
<code>removeXMarks</code>	boolean	True = render X marks along line (removal tools)
<code>dashArray</code>	number[]?	Optional line dash pattern (e.g. [12,6])
<code>calloutText</code>	string?	Single-line callout for transfer tools
<code>calloutLines</code>	string[]?	Multi-line callout for T&E, buried splice, etc.
<code>trenchDetail</code>	object?	Trench dimension box fields: Length, Width, Depth, etc.

`resolveSmartUnit(category, symbolKey, attributes)` — Primary Resolver

Converts the raw `mapping()` output into a normalized billing result consumed by `storeUnitData()`:

Field	Description
<code>unit_code</code>	Primary billing unit code (string)
<code>qty</code>	Quantity as number (no <code>Math.round</code> — decimals preserved)
<code>unit</code>	"EA" "FT" "HRS" "CY" etc.
<code>extraUnits[]</code>	Additional billing lines: { <code>unit_code</code> , <code>unit</code> , <code>qty</code> }
<code>color</code>	Hex color string
<code>label</code>	Canvas label text
<code>removeXMarks</code>	Boolean for X-mark rendering on removals
<code>calloutText</code> / <code>calloutLines</code>	Callout box content
<code>dashArray</code>	Line dash array
<code>trenchDetail</code>	Trench dimension data object

`storeUnitData(fabricObject, resolved)`

- Assigns a `__uid` to every canvas object
- Stores billing entries in `state.unitMap` (JS Map keyed by uid)

- Handles Format A: unit_code + extraUnits[] (most tools)
- Handles Format B: units[] array with code key (cable tools)
- Filters out zero-qty entries (e.g. occupied conduit adder when unchecked)

updateBillableUnits() — Panel Renderer

- Cross-references state.unitMap against live canvas objects
- Purges stale entries for deleted objects
- Aggregates all entries, renders HTML in the floating Billable Units panel

7. Complete Tool Inventory & Billing Logic

AERIAL TAB

POLE (aerial/POLE) — Shape: (X) circle with X

Status	Billing Units Generated
EXISTING	No billing. Canvas label = A-tag only (A-xxxxx)
NEW	POLE WOOD <=40ft OR >40-55ft (LABOR, EA) POLE {height}-{class} DF (MATERIAL, EA) + GPS COORDINATES if GPS entered + POLE WOOD <=40ft HAND SET if Hand Set checked
REMOVE	RMV POLE - CTL OWNED or RMV POLE - FOREIGN (LABOR, EA) SELECT BACKFILL 3/4in MINUS ABC (MATERIAL, 0.5 CY)

Rules: A-Tag auto-prefixes "A-" if not present. Height/Class/Hand Set/GPS only appear when Status=NEW. Owner dropdown only appears when Status=REMOVE. Hand Set default = unchecked. SELECT BACKFILL qty = 0.5 CY (not 1).

STRAND 10M (aerial/STRAND_10M) — dashed line

Status	Code
NEW	STRAND 10M (LABOR, FT)
REMOVE	RMV STRAND - BARE (LABOR, FT) + X marks on line

Note: 6M strand removed from all logic per contract update. Only 10M used.

DE/RELASH (aerial/DE_RE) — purple line

Config	Code
Always	AERIAL DELASH - RELASH (LABOR, FT) — Color: #A020F0

RE-TENSION (aerial/RE_TENSION) — arrow

Config	Code
Always	AERIAL SPAN RE-TENSION (LABOR, EA)

DOWN GUY (aerial/DOWN_GUY) — diagonal arrow

Status	Billing Units
NEW	DOWN GUY (LABOR, EA) DOWN GUY {10M or 16M} (MATERIAL, EA) DOWN GUY GUARD (MATERIAL, EA)
REMOVE	RMV POLE ANCHOR ROD (LABOR, EA)

ANCHOR (aerial/ANCHOR) — filled diamond

Status	Billing Units
NEW	ANCHOR SCREW {size} OR ANCHOR EXPANSION / BUST {size} (MATERIAL, EA)
REMOVE	RMV POLE ANCHOR ROD (LABOR, EA)

UNDERGROUND TAB

HH — Handhole (underground/HH) — Shape: rectangle with "HH"

Status	Billing Units
EXISTING	No billing
NEW	HH {size} (LABOR, EA) — sizes: 17x30x24, 30x48x36, 24x36x24, 48x48x48
REMOVE	RMV HH (LABOR, EA)

Rule: Canvas label always shows "HH" inside rectangle. Size NOT rendered on as-built.

MH — Manhole (underground/MH) — Shape: circle with "MH"

Status	Billing Units
EXISTING	No billing (unless checkboxes checked)
NEW	MH {size}in (LABOR, EA) — sizes: 48x72x84, 72x72x72, 72x144x84
REMOVE	RMV MH (LABOR, EA)
+ Crew entered MH (checkbox)	VAULT - TEST PURGE & VENTILATE (LABOR, EA)
+ Dewatering required (checkbox)	DEWATERING BASIC (LABOR, EA)

Rules: Both checkboxes default UNCHECKED — do NOT auto-check. MH size NOT rendered on canvas.

PEDESTAL (underground/PEDESTAL) — Shape: square with X

Status	Billing Units
EXISTING	No billing

Status	Billing Units
NEW	PED {size} {mount} (LABOR, EA) or PED FIB {size} {mount} for Fiber type
REMOVE	RMV PEDESTAL (LABOR, EA)

TRENCH (underground/TRENCH) — brown line

Configuration	Billing Units
No conduit	TRENCH {cover depth} CVR <=2.5in (LABOR, FT)
With conduit — 1st run	TRENCH {depth} CVR STICK {<=2.5in or 3-6in} (LABOR, FT)
With conduit — each additional run	TRENCH {depth} CVR STICK {bracket} ADDL (LABOR, FT × extra count)
Conduit material	CONDUIT {size} {schedule} 20ft GRY (MATERIAL, EA — sticks = ceil(footage/20))
Sweeps 90°	SWEEP {size} 90x36 PLASTIC BELL (MATERIAL, EA)
Sweeps 45°	SWEEP {size} 45x36 PLASTIC BELL (MATERIAL, EA)
Sweeps 22.5°	SWEEP {size} 22.5x60 C PLASTIC BELL (MATERIAL, EA)
Trench detail box rendered on canvas: Length × Width × Depth, Backfill, Spoils, Cold Mix.	

BORE (underground/BORE) — brown dashed line

Config	Code
Always	SPECIAL QUOTE - BORE - LE (LABOR, FT). Canvas label: BORE {footage}ft

SPLICE PIT (underground/SPLICE_PIT) — dashed square with crosshatch

Always NEW (no existing/remove variant).

Size	Unit Code
Small	PIT SMALL 6x4x5ft (LABOR, EA)
Medium	PIT MEDIUM 12x6x5ft (LABOR, EA)
Large	PIT LARGE 12x12x6ft (LABOR, EA)

POTHOLING (underground/POTHOLE) — bold circle with "P" (brown #8B4513)

Configuration	Billing Units
Hard Surface	POTHOLE HARD SURFACE (LABOR, EA)
Soft Surface	POTHOLE SOFT SURFACE (LABOR, EA)
+ Select Backfill checked	SELECT BACKFILL (MATERIAL, EA × qty)
+ Cold Mix checked (hard only)	ASPHALT TEMPORARY COLD PATCH (MATERIAL, SQFT)

MH GRADE ADJ (underground/MH_GRADE_ADJ)

Config	Code
Always	HH GRADE ADJUSTMENT (LABOR, EA × qty). Renders as callout arrow.

CABLE TAB

All cable tools render their abbreviation label in the center of the drawn line (e.g. "ASW", "BSW", "48F", "100C").

COPPER CABLE (cable/COPPER_CABLE)

Configuration	Billing Units
New Aerial (LASH)	COP CABLE {size} ANMW (MATERIAL, FT) LASH COPPER ≤400pr OR >400pr (LABOR, FT)
New Aerial (OVERLASH)	Same material + OVERLASH COPPER ≤400pr OR >400pr (LABOR, FT)
New Underground	Same material + COPPER CABLE IN CONDUIT ≤400pr OR >400pr (LABOR, FT)
Remove Aerial	RMV AERIAL COPPER ≤400pr OR >400pr (LABOR, FT)
Remove Underground	RMV UG COPPER ≤400pr OR >400pr (LABOR, FT)
+ Occupied Conduit checked	OCCUPIED CONDUIT ADDER (LABOR, FT) — auto-uses same footage

FIBER CABLE (cable/FIBER_CABLE)

Configuration	Billing Units
New Aerial (LASH)	FIB {count} 1JKT 1ARMOR LT (MATERIAL, FT) LASH FIBER (LABOR, FT)
New Aerial (OVERLASH)	Same material + OVERLASH FIBER (LABOR, FT)
New Aerial (ADSS)	Same material + LASH FIBER (ADSS billed same per contract)
New Underground	Same material + FIBER PL IN CONDUIT (LABOR, FT)
Remove Aerial	RMV AERIAL FIBER (LABOR, FT)
Remove Underground	RMV FIBER FROM CONDUIT (LABOR, FT)
+ Occupied Conduit checked	OCCUPIED CONDUIT ADDER (LABOR, FT) — auto-uses same footage

ASW — Aerial Service Wire (cable/ASW)

Configuration	Billing Units
New	SVC DROP AER COP {size} (MATERIAL, FT) AERIAL SVC DROP (LABOR, FT)
Sizes	2pr-22g SJSS, 3pr-22g SHD SS, 6pr-22g RND SJSS, 6pr-22g SHD SS

BSW — Buried Service Wire (cable/BSW)

Configuration	Billing Units
Direct Buried	SVC DROP AER COP {size} (MATERIAL, FT) GROUND LAY SVC DROP (LABOR, FT)
In Conduit	Same + OCCUPIED CONDUIT ADDER (LABOR, FT — auto at same footage)

Removal Cable Tools

Tool	Unit Code
RMV FIBER	RMV AERIAL FIBER or RMV FIBER FROM CONDUIT (per environment)
RMV COPPER	RMV AERIAL COPPER <=400pr / >400pr or UG equivalents
RMV ASW	RMV AERIAL SVC DROP (LABOR, FT)
RMV BSW	RMV SVC DROP FROM CONDUIT or RMV GROUND LAY SVC DROP

All removals: green #00AA00, X marks rendered intermittently along the line.

SPLICING TAB

SPLICE WIZARD (splicing/SPLICE_WIZARD) — diamond shape

Builds a complete splice billing package:

- Setup/Teardown: SPL SETUP-TEARDOWN {LOCATION}-{FIB/COP} (LABOR, EA) — locations: Aerial, Handhole, Pedestal-Cabinet, Vault-MH, Building
- Fiber count: SPLICE FIBER FUSION {range} (LABOR, EA) — ranges: 1-4, 5-12, 13-24, 25-48, 49-144, 145-288, 289-432, 433-864
- Copper: SPLICE COPPER MECHANICAL 1-25pr / 26-100pr / MODULE <=300pr (LABOR, EA)
- Cable prep: PREP FIBER CABLE IN HOUSING or copper equivalent (LABOR, EA)
- Case material: user-entered case code (MATERIAL, EA)

RISER (splicing/RISER) — [-| shape

Status	Billing Units
NEW	RISER PIPE POLE {size} {PLASTIC/STEEL} (MATERIAL, EA) POLE RISER - STANDOFF (LABOR, EA) + RISER U-GUARD {size} {material} if selected
REMOVE	RMV POLE RISER OR U-GUARD (LABOR, EA)
TRANSFER	REARRANGE XFER POLE RISER (LABOR, EA)

GROUND ROD (splicing/GROUND_ROD) — [GND] shape

Only rendered when Status = NEW on a pole. Types: GROUND ROD 5/8in-8ft, 5/8in-20ft, 1/2in-5ft (MATERIAL, EA)

BURIED SPLICE (splicing/BURIED_SPLICE) — callout box

- CASE BUR COP SOFT {kit} (MATERIAL, EA)
- CASE COPPER BURIED SOFT (LABOR, EA)

- LOCATE MARKER BALL (LABOR, EA)
- SPLICE COPPER MODULE {size} (LABOR, PR × pair count)
- SPL SETUP-TEARDOWN {location}-COP (LABOR, EA)

XFERS TAB (Transfers — all blue #0000FF)

A unified XFERS button in the Aerial tab opens a checklist modal. Each transfer renders as a draggable arrow + bordered callout box.

Tool	Unit Code
XFER POLE ATTACH	REARRANGE XFER POLE ATTACHMENT (LABOR, EA)
XFER POLE ATTACH ADDL	REARRANGE XFER POLE ATTACHMENT ADDL (LABOR, EA)
XFER SVC DROP	REARRANGE XFER AERIAL SVC DROP ADDL (LABOR, EA)
XFER SMALL FAC	REARRANGE XFER POLE MOUNT SMALL FAC (LABOR, EA)
XFER LARGE FAC	REARRANGE XFER POLE MOUNT LARGE FAC (LABOR, EA)
XFER POLE TAG (Copper)	TAG AER COP CABLE (LABOR, EA)
XFER POLE TAG (Fiber)	TAG AER FIB CABLE (LABOR, EA)

MISC TAB (callout boxes, no canvas render)

Tool	Billing Unit	Unit
T&E	Crew label (e.g. T&E 3 MAN AERIAL CREW)	HRS
DOWNTIME	DOWNTIME - CAPITAL PROJECT	HRS
SPLICER FIBER	SPLICER - FIBER	HRS
SPLICER COPPER	SPLICER - COPPER	HRS
EMERGENCY TRAVEL	EMERGENCY TRAVEL TIME	HRS
VAC TRUCK	No billing — vendor invoice callout	—

T&E; and DOWNTIME: rate auto-extracted from crew label (e.g. \$320/hr) → dollar amount calculated and shown in callout box with required justification.

TOOLS TAB (Basic Drawing — No Billing)

Tool	Behavior
LINE	Straight line — click and drag
ARROW	Line with arrowhead
RECT	Rectangle
CIRCLE	Ellipse

Tool	Behavior
CALLOUT	Speech bubble callout box
TEXT	Editable i-text object
FREEHAND	Free draw (Fabric.js pencil brush)

8. Map & Background Image Features

Google Maps Integration

- Modal with satellite map view
- Auto-centers on user geolocation when opened
- "(O) MY LOCATION" button re-centers + drops blue dot marker
- Capture → takes static satellite screenshot → places on canvas as selectable, resizable image

Map Image Behavior on Canvas

- Placed at canvas origin (0,0), scaled to fill canvas
- Fully selectable and resizable — has Fabric.js resize handles
- Tagged with nscData = { type: "MAP_BACKGROUND" }

Right-Click Context Menu (on map/photo images)

Option	Behavior
Copy / Paste	Clone the image, offset by 20px, place on canvas
Send to Back / Bring to Front	Change z-order
Delete	Remove from canvas
Flatten to Background	Locks image in place, non-interactive, sent to back. Drawn objects stay editable on top.
Unflatten (Re-enable Resize)	Restores full resize/move handles for repositioning

Photo Import

- File input accepts images, placed at canvas center-left
- Tagged with nscData = { type: "PHOTO_IMPORT" }
- Same right-click context menu as map background

9. Key JS Functions Reference

Function	File	Purpose
safelnit()	app.js	App entry point — wraps init()
sizeCanvas()	app.js	Resize canvas to fill available space, set CSS vars

Function	File	Purpose
drawGrid()	app.js	Render dot-grid background pattern
initTabs()	app.js	Wire tab nav clicks, auto-open Tools tab on load
setTool(name)	app.js	Activate a basic drawing tool
setTelecomTool(cat, key, type)	app.js	Activate a billing tool by category + key
bindTeleButtons()	app.js	Wire all sidebar buttons after DOM injection
showModal(key, attrs, onConfirm, onDiscard)	app.js	Render premium dark-glass modal
syncAllFields()	app.js	Collect all modal field values on confirm
refreshConditionalFields()	app.js	Show/hide fields based on showWhen rules
updateSelectionUI()	app.js	Update style strip when object selected
storeUnitData(obj, resolved)	app.js	Write billing data to state.unitMap
updateBillableUnits()	app.js	Re-render Billable Units floating panel
pushHistory()	app.js	Snapshot canvas for undo (max 30 states)
zoomCanvas(factor)	app.js	Zoom in/out by factor
resetZoom()	app.js	Reset to 100% zoom
resolveSmartUnit(cat, key, attrs)	units.js	PRIMARY billing resolver — returns normalized result
resolveUnit(key, attrs)	units.js	Legacy wrapper with auto category detection

10. State Object

Property	Type	Description
activeTool	string null	Current drawing tool (e.g. "LINE", "XFERS")
activeCategory	string null	"aerial" "underground" "cable" "splicing" "xfers" "misc"
activeKey	string null	Symbol key within category (e.g. "POLE", "FIBER_CABLE")
unitMap	Map	uid → [{ unit_code, desc, unit, qty }] billing entries
history	array	Array of JSON canvas snapshots for undo/redo
historyIndex	number	Current position in history stack
zoom	number	Current zoom level (1 = 100%)
pan	object	{ x, y } current pan offset

11. Business Rules & Contract Logic

Rule	Detail
A-Tag prefix	Always auto-prefixes "A-" if not already present (e.g. "12345" → "A-12345")
SELECT BACKFILL qty	0.5 CY (not 1) — Math.round() intentionally removed from billing
Strand	Only 10M strand. 6M removed from all logic per contract update.
No COAX	Removed from entire app — not a Lumen contract item
No Arborist in UI	Subcontractor-only unit — not a foreman tool
MH/HH/Ped size on canvas	NOT shown unless Status = NEW
Removal lines	Green (#00AA00) with X marks intermittently along the path
XFERS button routing	Routes to setTool("XFERS") BEFORE activeCategory gate — explicit bug fix
Occupied Conduit Adder	Bills at SAME footage as cable run — no separate footage field
Splice Pits	Always NEW — no existing/remove variant
Ground Rod	Only shown when Status = NEW on pole tool
RMV BURIED FAC location	Lives in UG tab. RMV ANCHOR lives in Aerial tab. Removals tab removed.
MH crew checkboxes	Both default UNCHECKED — never auto-check
Hand Set default	Unchecked — never auto-check

12. Smartsheet Integration & Cloudflare Worker

Cloudflare Worker

URL: <https://nsc-smartapp.williamkeesee06.workers.dev>

The worker proxies Smartsheet API requests because Smartsheet does not allow direct browser calls (CORS).

Request: ?wo={workOrderNumber} → returns job JSON.

Smartsheet Column ID Mapping

Field	Column ID
Work Order	4680657223346052
Supervisor	8776041144995716
Foreman	4146149265985412
Address	7094520111753092
City	1465020577539972
Zip	4822376816371588
Job Status	4721327342413700

Field	Column ID
TC Required	6789809108373380
Notes	6458790863759236

Job Lookup Response Fields

- supervisor, foreman, address, city, zip, status, tcRequired, notes
- Auto-geocodes address → centers Google Map + drops orange arrow marker
- Status indicator: amber <<>> SEARCHING → green OK or red X NOT FOUND

13. UI Theme

Main UI — Clean White/Blue

Property	Value
Font	Inter
Page background	#f0f2f5
Surface	#ffffff
Header	White, 4px bottom border #0052cc
Tab strip	#0052cc sidebar, active tab = white bg + blue text
Accent / Primary	#0052cc
Tool buttons	#f8fafc bg, border #cbd5e1, hover: blue
Style strip	White bg, 2px bottom border #0052cc
Canvas	White with dot-grid background

Modal Theme — Dark Glass

Property	Value
Background	linear-gradient(160deg, #0d1421, #111827)
Border	rgba(56,189,248,0.25) — neon blue
Title font	Rajdhani, 17px, letter-spacing 0.14em
Neon accent	#38bdf8
CONFIRM button	Neon blue gradient with glow box-shadow
DISCARD button	Ghost metallic
Inputs	Dark glass rgba(255,255,255,0.04) bg, focus neon glow

14. Deployment

Property	Value
Hosting	Vercel (static)
Live URL	https://www.perplexity.ai/computer/a/north-sky-as-built-tool-pFRkJnPfTcK.qFJkDSEXIQ
Asset ID	a4546426-73df-4dc2-bea8-52640d211795
Deploy command	deploy_website with should_validate=false
Entry point	index.html
Project path	/home/user/workspace/nsc_project

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